

‘city’ as the most probable next token, though it may also respond with ‘capital of Karnataka’, ‘Silicon Valley of India’, or something poetic like ‘romantic’—all based on probabilities.

This means LLMs don’t truly understand language as humans do. They generate responses based on statistical token relationships. Even though their creators admit they function like black boxes, nobody fully understands how certain outputs are generated.

The hallucination challenge

One major LLM challenge is hallucination. Since these models rely on probabilities, they sometimes generate information that sounds correct but is factually wrong. Early GitHub Copilot versions often generated non-existent code snippets, leaving developers wondering when such code was ever published.

Rule-based systems can help here. By adding symbolic layers enforcing rules and constraints, we can reduce hallucinations. This combination of statistical and rule-based reasoning is the essence of hybrid reasoning.

How hybrid reasoning functions

From my perspective, hybrid reasoning combines the structure of symbolic AI with the flexibility of statistical AI. Symbolic AI provides rules and constraints while statistical AI offers adaptability. Together, they deliver more accurate, trustworthy, and contextually appropriate outputs.

Hybrid reasoning involves two components.

- *Fast thinking:* Rule-based symbolic systems that execute immediately since rules are either satisfied or not.
- *Slow thinking:* Statistical reasoning that breaks complex queries into sub-problems, analyses them, and combines the results.

When receiving a query, the system determines whether both thinking types are required and then combines them for the final response.

Real-world applications

Let me share practical examples of the

value of hybrid reasoning.

Consider working with large legacy codebases, where every change risks breaking something else. Relying solely on LLMs may generate statistically probable but contextually wrong suggestions. However, introducing symbolic reasoning through rules and knowledge graphs provides constraints that improve accuracy.

Another example involves sensitive queries. Certain LLMs like DeepSeek avoid controversial political questions. Ask directly about a political leader’s failures, and it refuses to answer. But with hybrid reasoning, you can design workflows that reframe queries to draw factual responses without breaking rules.

I believe hybrid reasoning’s value lies in integrating symbolic reasoning, statistical reasoning, and retrieval techniques like RAG, making outputs both accurate and contextually aligned with user needs.

Key benefits

In my experience, hybrid reasoning offers several clear advantages.

- *Improved accuracy:* Combining symbolic and statistical approaches reduces hallucinations and produces more factually correct answers.
- *Increased trust:* Users understand how outputs are derived because symbolic reasoning adds transparency, building system confidence.
- *Greater efficiency:* Instead of endless prompt iterations, hybrid reasoning achieves better responses in fewer attempts.
- *Enhanced flexibility:* The system adapts to different contexts, whether handling sensitive political topics or technical legacy system challenges.

Challenges and limitations

Hybrid reasoning isn’t without challenges though.

- *Increased complexity:* Adding more system layers means symbolic reasoning, statistical reasoning, and RAG must work together seamlessly.
- *Data quality dependency:* Poor input

data produces poor output—the classic ‘garbage in, garbage out’ principle.

- *Higher computational costs:* More layers require more processing, increasing query costs.

These challenges make optimisation and fine-tuning essential for real-world applications.

The future

The field is evolving rapidly in several promising directions.

- *Agent-based AI:* Specialised bots handling specific tasks more efficiently.
- *Neuro-symbolic AI:* Stronger synergy between rule-based and statistical systems.
- *Quantum learning:* Early explorations combining quantum computing with machine learning.
- *Ethics and governance:* Stronger frameworks ensuring responsible AI use.

I believe hybrid reasoning isn’t just another buzzword—it’s a meaningful step forward in AI evolution. By combining symbolic and statistical reasoning strengths, it enables building more accurate, trustworthy, and flexible systems.

Whether solving coding challenges, reducing hallucinations, or addressing sensitive queries, hybrid reasoning provides a balanced path. It reflects how humans often think—sometimes relying on intuition and rules, sometimes pausing to weigh probabilities and scenarios.

For me, the excitement lies not just in the technology but in the possibilities it offers. If a system designed to avoid controversial answers can still provide nuanced insights through hybrid reasoning, imagine its potential for enterprises, legacy systems, and daily workflows. This is why I believe hybrid reasoning deserves understanding and adoption as AI innovation advances. **END** 🐧

*The article is based on the keynote address titled ‘Harnessing the Power of Hybrid Reasoning’ given at AI DevCon by **Laxminarayan Chandrashekar**, technical architect at Siemens Technology. It has been transcribed and curated by **Vidushi Saxena**, journalist at OSFY.*